

The saturated dual resolvent

A specialist dossier on the critical bordered block $R^\dagger$

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Abstract

On a family of explicit finite discrete measures, a particle-hole duality identifies a two-measure subordination inequality with the spectral condition $R \succ \frac{1}{2}I$ on a dual hole kernel. A rank-one border of that kernel produces a single object $R^\dagger$ whose positivity $R^\dagger \succ \frac{1}{2}I$ is equivalent to the original inequality *and* a terminal budget condition, and whose critical three-by-three Schur block carries both. A constrained equilibrium problem on the same dual weight places the binding pair of nodes astride a saturated hard-edge block. All identities below are finite linear algebra; all numerical values are a frozen census on a finite family. No asymptotic theorem is proved. The one open question is stated as Problem 1. This document is a problem statement for specialists, not a claim, and nothing in it is evidence for or against the Riemann hypothesis.

Status and grades. Every assertion is marked EXACT (finite identity, including rational arithmetic), THEOREM-GRADE (finite-matrix theorem on measured inputs), CENSUS (measured on a frozen finite family; no extrapolation), or HYPOTHESIS (unproved analytic statement). The construction of the underlying measures (μ, ν) is given in full in the companion note [4]; the present dossier does not repeat it. Uvarov-transform language, the discrete-hard-edge placement, and the Lubinsky-class exclusion are recorded there (§frozen-placement, §frozen-dualhole) and are cited, not duplicated. Machine records: sealed probes `borodin_dual_hole_probe.py` (round 356), `schur_wronskian_dual_probe.py` (359), `critical_saturation_probe.py` (360), `augmented_borodin_duality_probe.py` (362).

1 Setup

Fix a prime power z (the window anchor). A triangular (“tent”) test function of half-width Δ , determined by z and a fixed oversampling parameter, samples the von Mangoldt comb and an archimedean kernel, producing an even trigonometric polynomial f — the symbol of a circulant. Folding the symbol onto $[-1, 1]$ by $x_j = \cos \theta_j$ yields a signed discrete measure on the cosine grid, split by sign into two positive measures μ and ν with disjoint supports in $[-1, 1)$ and S atoms in total [4, construction]. Write $N = \lceil S/2 \rceil$ and $Y = \text{supp } \nu$. Across the whole family the grid is uncultured, so $S = 2N - 1$ [exact]: half-filling is Borodin’s complementary rank condition, not an extra hypothesis.

The running instance is $z = 16$: $S = 367$, $N = 184$, $|Y| = 104$. The binding pair of ν -atoms sits at grid (fold) indices $(2, 4)$, the two rightmost odd nodes of the cosine lattice. The inequality under study, denoted L^* , asks

$$\int p^2 d\nu < \int p^2 d\mu \quad \text{for every real polynomial } p \neq 0 \text{ with } \deg p < N.$$

Equivalently, the ν -sampling matrix E_N in the μ -orthonormal frame satisfies $\lambda_{\max}(E_N) < 1$. An augmented form $L^\dagger$ adjoins a rank-one border (a budget-normalized smooth test functional)

and asks $\lambda_{\max}(\mathbf{E}_N + bb^T/B) < 1$. Both statements are open for the unbounded family. On every tested window the inequalities hold, with spectral margins decaying as a power of N ; structurally destroyed comparison inputs fail already at a small fraction of the critical degree [4]. No statement here claims either inequality.

2 Definition of $R^\dagger$

Lift (μ, ν) to the positive measure $\eta = \mu + \nu$ with weights $u_j = |\tilde{w}_j|$ on the union $X = \text{supp } \eta$. Let P_X be the monic node polynomial of X . Borodin’s reciprocal weight [1] is $u_j^\vee = 1/(u_j P_X'(x_j)^2)$. On this cosine grid the dual weight has the closed form

$$u_j^\vee \propto \frac{c_j(1-x_j)}{|f(\theta_j)|}, \quad (1)$$

where c_j is the folding weight ($c_j = \frac{1}{2}$ at the left endpoint, else 1): the original weight carries $+\log|f|$, the dual weight $-\log|f|$ [exact, three evaluation routes]. Let $\Pi_{N-1}^{u^\vee}$ be the rank- $(N-1)$ Christoffel–Darboux projection kernel of $u^\vee$ on X , dressed by $\sqrt{u^\vee}$, and let R be its restriction to Y .

Let b be the border functional in the μ -orthonormal frame, $B > 0$ the budget, $\beta = b/\sqrt{B}$, and v the node-side image of the border under the μ -Christoffel–Darboux kernel ($v_j = \sqrt{\nu_j/B} \int K_\mu(y_j, t) d\sigma(t)$). Write $\gamma = \beta^T \beta$ and $\tilde{v} = Dv$, with D the diagonal sign matrix of P_X' .

Definition 1 (Bordered dual resolvent). *The augmented dual kernel is the $(|Y|+1) \times (|Y|+1)$ matrix*

$$R^\dagger = \begin{pmatrix} R^{-1} & \tilde{v} \\ \tilde{v}^T & 1 + \gamma \end{pmatrix}^{-1}. \quad (2)$$

The extra coordinate is the border; the leading principal block is a rank-one Sherman–Morrison deformation of R (Proposition 2 (A4) below).

Route (i) — adjoining a virtual atom and taking the $(S+1)$ -point Borodin kernel — is available on a single-atom toy border (half-filling shifts $S = 2N - 1 \mapsto S^\dagger = 2N$, dual rank $N - 1 \mapsto N$) and agrees with (2) up to sign conjugation [exact, rational arithmetic]. On the real windows the border is a signed smooth-comb measure with S atoms, every one of which collides with a grid node, and is not a point evaluation (representability spread 56.8 on the running instance). The bordered-resolvent form is therefore the general object.

3 Exact duality formulae

Write $\text{margin} = 1 - \lambda_{\max}(\mathbf{E}_N)$ and $\text{margin}^\dagger = 1 - \lambda_{\max}(\mathbf{E}_N + bb^T/B)$.

Proposition 1 (Particle–hole dictionary [1]). *At half-filling $S = 2N - 1$,*

$$L^* \iff \lambda_{\max}(\mathbf{E}_N) < 1 \iff R \succ \frac{1}{2}I,$$

with the spectral map $\text{margin} = 2 - 1/\lambda_{\min}(R)$ [theorem-grade; absolute deviation $\leq 9.9 \cdot 10^{-10}$ on 85 windows]. The L -ensemble transformation $Q = \mathbf{E}_N(I + \mathbf{E}_N)^{-1}$ is exact [rational arithmetic on a 5-atom model; double precision $\leq 7.6 \cdot 10^{-12}$ on 85 windows], and combined with Borodin’s complementation $\Pi_N^u = D(I - \Pi_{N-1}^{u^\vee})D$ yields $\mathbf{E}_N = D(R^{-1} - I)D$ on Y .

Live census: $\lambda_{\min}(R) - \frac{1}{2}$ ranges from $+4.2 \cdot 10^{-5}$ (running instance: $\lambda_{\min}(R) = 0.500041882$) down to $+5.9 \cdot 10^{-11}$ on the deepest tranche — all positive, no counterexample. Comparison inputs with destroyed source structure violate $R \succ \frac{1}{2}I$ structurally ($\lambda_{\min}(R) - \frac{1}{2} \in [-0.500, -0.083]$). Honest limit: at the deepest instances the double-precision resolution of the

dual route ($\sim 5 \cdot 10^{-10}$) exceeds the margins, so the hole coordinate is exact algebra, not a sharper certification instrument.

The pair identities $c = \varepsilon_1 \varepsilon_2 (R^{-1})_{12}$, $p = 2 - (R^{-1})_{11}$, $q = 2 - (R^{-1})_{22}$ identify the two-point Uvarov block of the companion note with the $(1, 2)$ principal minor of $2I - R^{-1}$ [theorem-grade]. The absolutely continuous hard-edge class is excluded at census grade: the rescaled pair positions $a_j = 2N^2(1 - y_j)$ are family-constant against $\pi^2 j^2/4$ on all 85 windows (maximum deviation $0.33/N$), while the kernel correlation ρ_K decays with fitted exponent 1.4222, contrary to Lubinsky's Bessel limit at fixed microscopic arguments [3], [4, §frozen-placement].

Proposition 2 (Augmented dictionary). *Let $Z = \begin{pmatrix} R^{-1} & \tilde{v} \\ \tilde{v}^T & 1+\gamma \end{pmatrix}$, so $R^\dagger = Z^{-1}$, and write $M^\dagger = R^\dagger - \frac{1}{2}I$. The following are finite-matrix identities [theorem-grade; rational arithmetic on toys, graded double precision on 75 windows of the prime-generated family plus two character-twisted families of 42 windows each].*

- (A1) *The augmented node Gram $G^\dagger$ is the sign-conjugate of $(R^\dagger)^{-1} - I$: the particle-hole complementation extends by one row.*
- (A2) *$\text{margin}^\dagger = 2 - 1/\lambda_{\min}(R^\dagger)$ (deviation $\leq 1.1 \cdot 10^{-11}$ shallow, $\leq 5.5 \cdot 10^{-10}$ at the deepest computed row).*
- (A3) *$L^\dagger \iff R^\dagger \succ \frac{1}{2}I$, two-sided. Both truth directions occur on real windows: six character-twisted rungs with terminal ratio $q_N > 1$ have $\lambda_{\min}(R^\dagger) - \frac{1}{2} < 0$ (e.g. $-9.8 \cdot 10^{-5}$), exactly as the identity demands.*
- (A4) *Sherman–Morrison on the Y-block: $R_{YY}^\dagger = R + (R\tilde{v})(R\tilde{v})^T/(1 + \gamma - \tilde{v}^T R\tilde{v})$, and $R_{bb}^\dagger = 1/(1 + \gamma - \tilde{v}^T R\tilde{v})$ (deviation $\leq 1.2 \cdot 10^{-14}$ on every computed row, including the deepest). The border is a rank-one insertion at dual-resolvent level.*
- (A5) *$R^\dagger \succ \frac{1}{2}I$ if and only if $R \succ \frac{1}{2}I$ and $q^\dagger < 1$, where $q^\dagger = \gamma + v^T(I - E_N)^{-1}v$ is frame-invariant (equals the squared dual border norm $u^T H^{-1}u/B$) and coincides with the standing terminal coordinate. The dictionary $1 - q^\dagger = (5/7)(1 - q_N)/B$ holds to relative $6.7 \cdot 10^{-6}$.*
- (A6) *The Schur complement of $M^\dagger$ onto the border coordinate equals $(1 - q^\dagger)/(2(1 + q^\dagger))$ globally, and equals the same value inside the local 3×3 block of Proposition 3 (quotient $\leq 1.0 \cdot 10^{-11}$).*
- (A7) *Cauchy interlacing: $\lambda_k(R^\dagger) \leq \lambda_k(R) \leq \lambda_{k+1}(R^\dagger)$. The border can only lower the bottom of the dual spectrum ($\text{margin}^\dagger \leq \text{margin}$). On the running instance, $0.500041459 \leq 0.500041882$.*

A formal reconstruction, independent of the dual coordinate, is already machine-checked in Lean: $L^\dagger \iff L^* \wedge \text{Terminal}$ and $L^* \wedge \text{Terminal} \Rightarrow$ positivity of every bordered window matrix through the cap (RH/Augmented.lean, augmentedSubordination_iff_lstar_and_terminal; RH/Closure.lean, lstar_terminal_implies_master; axiom audit in RH/Audit.lean, consuming only the two named canonical holes). What is *not* formalized, and not proved analytically, is $R^\dagger \succ \frac{1}{2}I$ for the unbounded family. That is Problem 1.

4 The critical three-by-three block

Partition $Y = \{y_1, y_2\} \cup C$, with $\{y_1, y_2\}$ the binding pair (folds 2, 4) and C the rest of Y . Write $M = R - \frac{1}{2}I$ and S_N for the 2×2 Schur complement of M onto the pair. Then [exact]

$$M \succ 0 \iff M_{CC} \succ 0 \text{ and } S_N \succ 0,$$

and S_N is a ratio of Sylvester-bordered minors of $\det M$ (the rest block sits inside the border; a pair-only truncation is false by 4.07 nats on the running instance and exactly on the rational toy). The off-diagonal of the dual resolvent $A^{-1} = (I - 2R)^{-1}$ is the Casoratian of the two consecutive dual orthonormal polynomials, resolvent-dressed (discrete IKS identity) [theorem-grade]. The diagonal entries $(A^{-1})_{kk}$ are *not* Casoratian data: their cross share $(A^{-1})_{12}^2 / ((A^{-1})_{11}(A^{-1})_{22})$ has median 0.702 on 85 windows (range 0.46–0.84). No standalone explicit prefactor turns the Casoratian into $\det S_N$ at the sealed bars.

Proposition 3 (Binding). *On the 74 resolvable windows of the family $(\lambda_{\min}(R) - \frac{1}{2} > 10^{-9})$,*

$$\text{bind} = \frac{\lambda_{\min}(S_N)}{\lambda_{\min}(R) - \frac{1}{2}} \in [1.0003, 1.0605], \quad \text{median } 1.0058$$

[census]. *After the rank-one border, the 3×3 Schur block $S_3^\dagger$ of $M^\dagger$ on (pair, border) satisfies*

$$\text{bind}^\dagger = \frac{\lambda_{\min}(S_3^\dagger)}{\lambda_{\min}(R^\dagger) - \frac{1}{2}} \in [1.0003, 1.0608], \quad \text{median } 1.0056$$

[census]. *The in-block border Schur equals the global border Schur of (A6). The rank-one deformation does not move the pair structure: on the running instance the cross share of $R^\dagger$ is 0.6990 against 0.6973 for R ; ladder medians 0.708 against 0.702.*

On the running instance the three eigenvalues of $S_3^\dagger$ are $(4.16 \cdot 10^{-5}, 4.90 \cdot 10^{-4}, 1.76 \cdot 10^{-2})$: the bottom mode is the L^* pair channel, the top mode is the terminal border channel. The critical eigenvector's border mass is $2.5 \cdot 10^{-5}$ there, and $5.8 \cdot 10^{-4}$ at the near-terminal window where $\text{margin}^\dagger / \text{margin} = 0.8716$. The giant operator behaves, at the critical point, like this 3×3 block.

5 The pair straddles the saturation edge

Two independent finite objects describe the dual ensemble's occupation of the cosine grid.

Exact occupation. Let $o_\eta(x_j)$ (resp. $o_{\text{dual}}(x_j)$) be the diagonal of the rank- N η -projection (resp. the rank- $(N-1)$ dual-hole projection). Borodin complementation on the diagonal gives

$$o_\eta(x_j) + o_{\text{dual}}(x_j) = 1 \tag{3}$$

at every union node, with trace ward $\sum_j o_{\text{dual}}(x_j) = N - 1$ [exact; rational arithmetic on the toy, $\leq 2.2 \cdot 10^{-11}$ live]. This is the finite counterpart of an equilibrium density versus its upper constraint, with no solver and no asymptotics. On the prime-generated family, fold 1 (the rightmost node) is the unique occupation anomaly: $o_{\text{dual}}(x_1)$ lies 0.25–0.33 below the bulk median on all 85 windows, while the pair folds 2 and 4 are flat (deviation ≤ 0.10).

Obstacle problem. Minimize the discrete energy

$$E(m) = \sum_{j \neq k} m_j m_k \log \frac{1}{|x_j - x_k|} + \sum_j m_j (-\log u_j^\vee)$$

over $0 \leq m_j \leq 1$, $\sum m_j = N - 1$. The box constraint is the Baik–Kriecherbauer–McLaughlin–Miller upper constraint [2]; dropping it piles mass ≥ 92 times the cap onto single nodes. A deterministic projected first-order solver (FISTA plus exact box-simplex water-filling, sealed iteration counts, KKT partition gates) produces, on four sealed instances including the running instance, a dual-void block equal to folds $\{1, 2, 3\}$ [census]. By (3) this is the primal saturated block at the right hard edge.

Proposition 4 (Straddle, census). *The binding pair (2, 4) straddles the saturation-block edge 3 | 4: fold 2 lies inside the saturated zone, fold 4 outside. Auxiliary character-twisted families on the same grid have a different partition — they occupy fold 1, with no fold-1 occupation anomaly on 0/42 + 0/42 windows — and a sign-scrambled comparison input is disordered at the edge while failing $R_{CC} \succ \frac{1}{2}I$ ($\lambda_{\min}(R_{CC}) - \frac{1}{2} = -0.4962$) with the pair block itself still positive. The identity chain is world-blind algebra; the partition, and the positivity, are not.*

The asymptotic equilibrium-measure theorem (BKMM band and saturation limit shape for the field $-\log u^\vee$) is not proved [hypothesis, open]. The finite partition is a measurement about this family.

6 The observed exponent

On the 57 calibration windows a Theil–Sen fit of $\log \lambda_{\min}(S_N)$ has slope -3.334 , hitting the standing margin exponent 3.332 at distance 0.002 and rejecting an alternative reading 1.65 at distance 1.684 [census]. The diagonal dual-resolvent entries carry the same exponent ($|(A^{-1})_{11}|$ slope $+3.365$, $|(A^{-1})_{22}| + 3.292$, $|(A^{-1})_{12}| + 3.316$). Weight-side Christoffel exponents (1.79 and 0.38 families) miss by a wide margin: they do not carry the resolvent diagonals.

A dimensionless combination $t_{\text{geo}} = (\lambda_{\min}(R) - \frac{1}{2}) \sqrt{(A^{-1})_{11}(A^{-1})_{22}}$ is margin-locked: slope -0.009 on the 57, range $[0.222, 0.287]$ on all 74 resolvable windows [census]. Exact algebra identifies t_{geo} with a function of $\det(S_N/\varepsilon)$ and the cross share; family-constancy is a local-parametrix-class census, not a Gamma-function derivation. The Gamma parametrix of [2, Thm 2.7, Rem 2.8] adjacent to a saturated band remains the natural analytic candidate [hypothesis, open].

An adjugate identity $\det S_N \cdot 4[(A^{-1})_{11}(A^{-1})_{22} - (A^{-1})_{12}^2] = 1$ [exact] implies that the wander of $\log \det S_N$ is the wander of $-\log((A^{-1})_{11}(A^{-1})_{22})$ (fine-structure correlation $+0.9835$), with the cross-share term five times subleading. The leading common wander lives in the diagonal resolvent data and cancels algebraically in the minor.

7 Relative error-operator form

Two constant-matrix targets fail the census and force a relative statement.

Rest block. Cauchy interlacing on the principal submatrix R_{CC} , and the Schur-complement eigenvalue inequality, give

$$\lambda_{\min}(R_{CC}) - \frac{1}{2} \geq c \left(\lambda_{\min}(R) - \frac{1}{2} \right) \quad \text{with } c = 1 \quad (4)$$

as a finite theorem, gated in both directions on the rational toy and on $74 + 84$ resolvable windows [theorem-grade]. On the prime-generated family the measured ratio lies in $[8.8, 54.2]$ (median 20.3). On the character-twisted families the ratio reaches the interlacing line: minima 1.0 on 11 of 84 windows. Any analytic rest-block clause is a theorem only at $c = 1$; a comfortable factor 8.8 does not transfer.

Leading coefficient. The schematic $S_N = N^{-\alpha}(M_0 + O(N^{-\varepsilon}))$ with M_0 a constant positive matrix is too rigid. The condition number $\kappa_S = \lambda_{\max}(S_N)/\lambda_{\min}(S_N)$ ranges in $[6.3, 30.5]$ (median 13.9), a range ratio $4.88 > 1.5$: the margin direction collapses (bind median 1.0058) but the second eigenvalue wanders by an $O(1)$ factor [census]. The honest form is projective: one controls $S_N/\lambda_{\min}(S_N)$ in a bounded set of positive-definite matrices, or equivalently the error operator $E = P_{\text{true}}P_{\text{model}}^{-1} - I$ of a local Riemann–Hilbert parametrix in a *relative* operator norm on the critical block, not as an additive constant remainder.

No bound on E is proved [open]. Equation (4) at $c = 1$ is the only theorem-grade error control presently available, and it is finite interlacing, not an asymptotic remainder.

8 Problem statement

Problem 1 (Critical positivity of the bordered dual resolvent). *Let (μ, ν) run over the admissible family of [4], with dual kernel R and bordered dual resolvent $R^\dagger$ as in Definition 1. Which theorem in the Riemann–Hilbert analysis of discrete orthogonal polynomials — in particular, which statement in the saturation regime of Baik–Kriecherbauer–McLaughlin–Miller [2, Thm 2.7, Remark 2.8] — guarantees*

$$R^\dagger \succ \frac{1}{2}I \tag{5}$$

uniformly for the unbounded family? Equivalently, given Proposition 3, which local parametrix at a hard edge adjacent to a saturated band guarantees that the critical 3×3 Schur block of $R^\dagger - \frac{1}{2}I$ (binding pair plus border) stays positive-definite?

The finite input to such a theorem is assembled above: reciprocal weight (1); occupation duality (3); a census-grade saturated block $\{1, 2, 3\}$ with the binding pair astride the edge $3 \mid 4$; observed rate $N^{-3.332}$ carried by the diagonal resolvent data; local-parametrix datum t_{geo} margin-locked in $[0.222, 0.287]$; error control in relative/projective form, with rest-block theorem only at $c = 1$. The AC/Bessel class is the wrong class [3]. A Gamma-class derivation at saturation adjacency is the standing analytic candidate and is open.

9 What a proof would give, and what it would not

A proof of (5) for the unbounded family would give, in order: $R^\dagger \succ \frac{1}{2}I \Rightarrow L^\dagger$ by Proposition 2(A3) [theorem-grade finite algebra]; $L^\dagger \Rightarrow L^* \wedge \text{Terminal}$ and $L^* \wedge \text{Terminal} \Rightarrow$ positivity of every bordered window matrix through the cap [Lean, sorry-free on the reconstruction layer, axiom audit `RH/Audit.lean`]. That is window-local master positivity. It is not an asymptotic law, not a bound on the Riemann ξ -function, and not a statement about the Riemann hypothesis in either direction.

What is on the table now is a finite census (at most 85 windows for the hole kernel, 75 plus one deep row for the border, four sealed obstacle instances for the partition) together with exact finite-matrix identities. No Riemann–Hilbert steepest-descent analysis has been carried out. The equilibrium-measure limit, the Gamma parametrix, and the error operator E are open. Comparison inputs show that the identities survive source destruction while positivity and the saturated partition do not — so an existence proof for (5) must consume the construction, not merely the dual algebra.

Table 1 records the running instance. Every number is a sealed probe record. Numbers not in those records are omitted.

Claim boundary. Research documentation. Not a proof of L^* , of $L^\dagger$, or of $R^\dagger \succ \frac{1}{2}I$ for the unbounded family. Not evidence for or against the Riemann hypothesis in either direction.

References

- [1] A. Borodin, *Duality of orthogonal polynomials on a finite set*, J. Statist. Phys. **109** (2002), 1109–1120.
- [2] J. Baik, T. Kriecherbauer, K. T.-R. McLaughlin and P. D. Miller, *Discrete Orthogonal Polynomials: Asymptotics and Applications*, Ann. of Math. Studies **164**, Princeton Univ. Press, 2007. Theorem 2.7 and Remark 2.8.

Table 1: Running instance $z = 16$ ($S = 367$, $N = 184$, pair folds $(2, 4)$). Dual quantities: round 356/359. Occupation and obstacle: round 360. Bordered quantities: round 362.

quantity	value
$\lambda_{\min}(R)$	0.500041882
$\lambda_{\min}(R^\dagger)$	0.500041459
margin / margin [†]	$1.6752 \cdot 10^{-4}$ / $1.6582 \cdot 10^{-4}$
bind / bind [†]	1.0029 / 1.0028
cross share / share [†]	0.6973 / 0.6990
t_{geo}	0.2646
o_{dual} at folds 1, 2, 4	0.1497, 0.5118, 0.5137
QP dual-void block (KKT gap)	$\{1, 2, 3\}$ (+0.092)
$q^\dagger$ / border Schur	0.933044 / $1.7319 \cdot 10^{-2}$

- [3] D. S. Lubinsky, *A new approach to universality at the edge of the spectrum*, Int. Math. Res. Papers (2008).
- [4] S. Hamann, *A strict polynomial subordination inequality for two explicit prime-generated discrete measures (open problem)*, TFPT research repository, `rh/problem/lstar_problem.tex`, August 2026.